

Week 2

W2.2- LLM Agent Basics

2026 Spring

[LLM Agents Foundation & Applications](#)

Dr. Yanjun Qi

20260115

Last Class

To incorporate human preferences:

- **RL algorithms** (PPO, GRPO, REINFORCE) explicitly maximize expected reward from a reward model – we normally call this group **RLHF**
- **Direct alignment methods** (DPO, IPO, KTO) optimize preference objectives without explicit reward modeling
- Though they can be shown to implicitly optimize an equivalent objective under certain assumptions

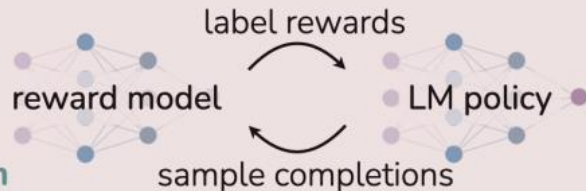
Reinforcement Learning from Human Feedback (RLHF)

x: "write me a poem about
the history of jazz"



preference data

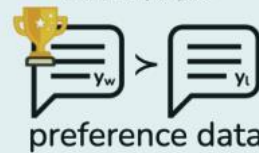
maximum
likelihood



reinforcement learning

Direct Preference Optimization (DPO)

x: "write me a poem about
the history of jazz"



preference data

maximum
likelihood



Text input → LLM → Text output

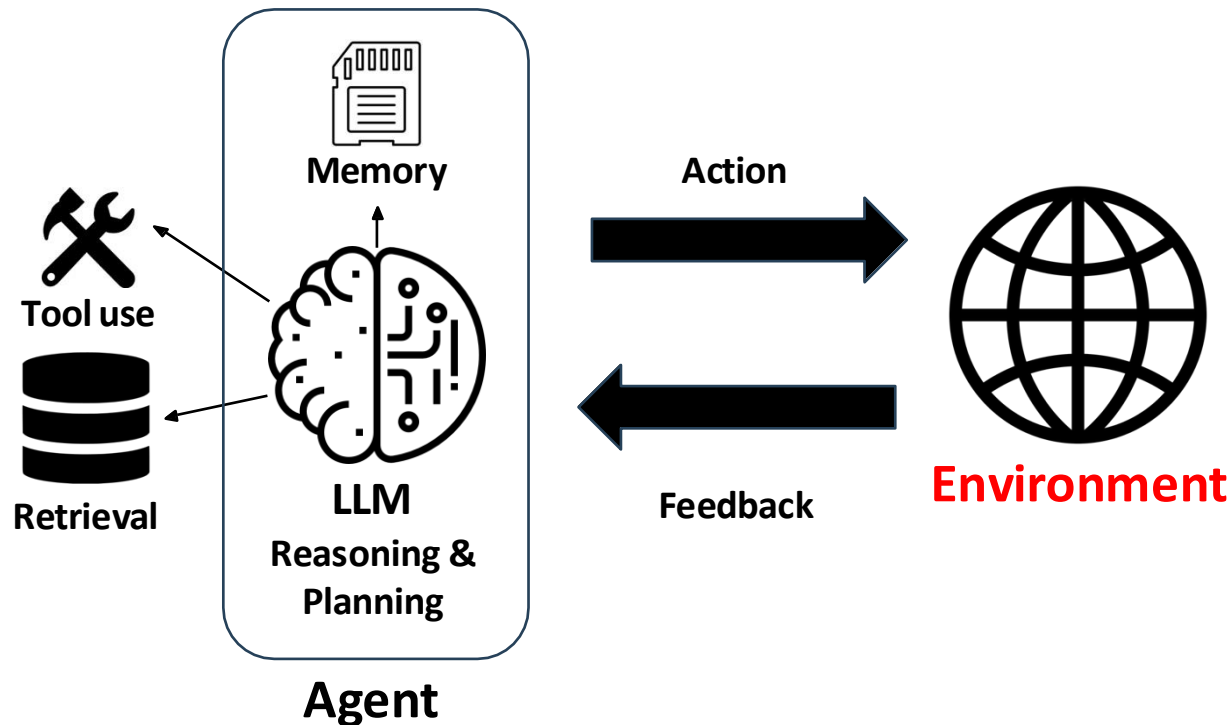
ranked by capabilities, sized by billion parameters used for training

CLICK LEGEND ITEMS TO FILTER

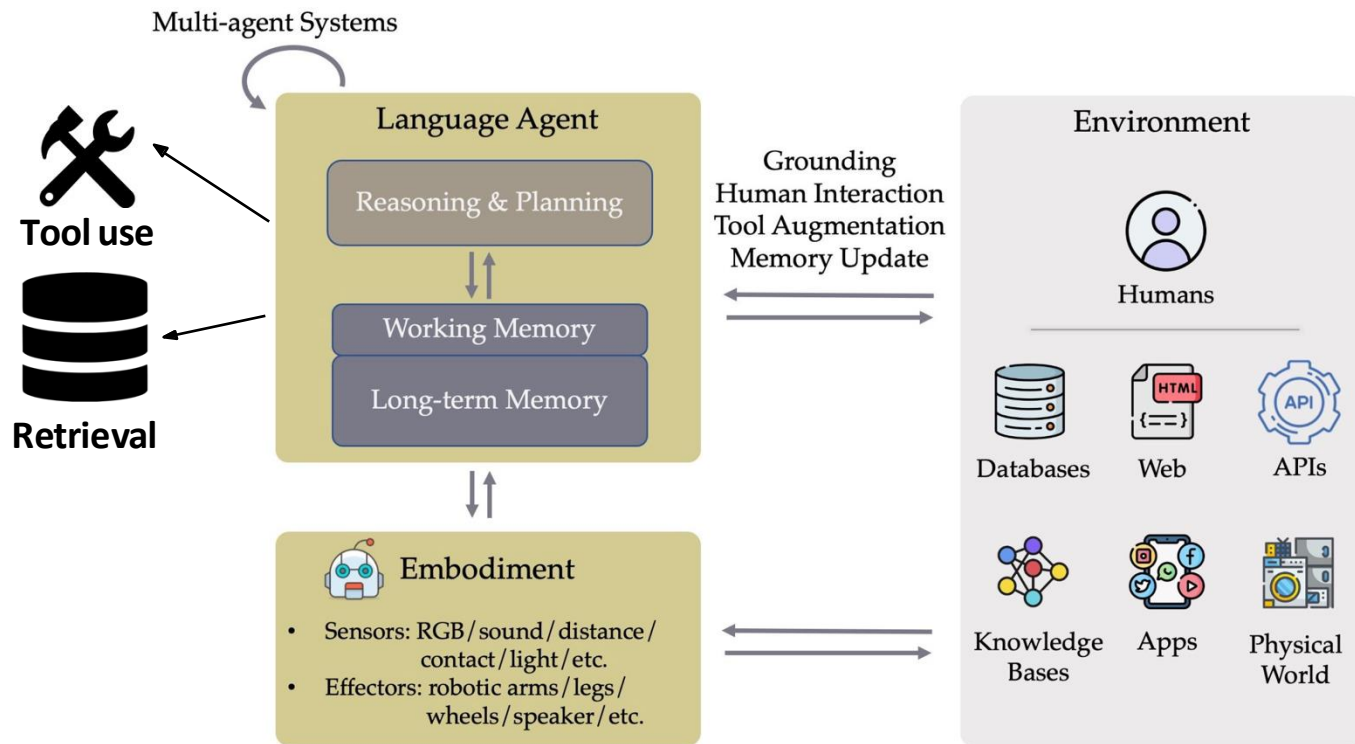


* = parameters undisclosed // source: LifeArchitect // data

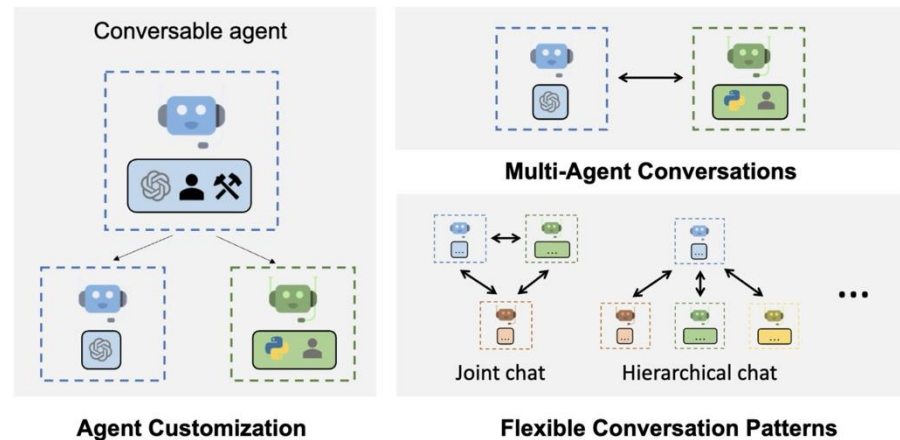
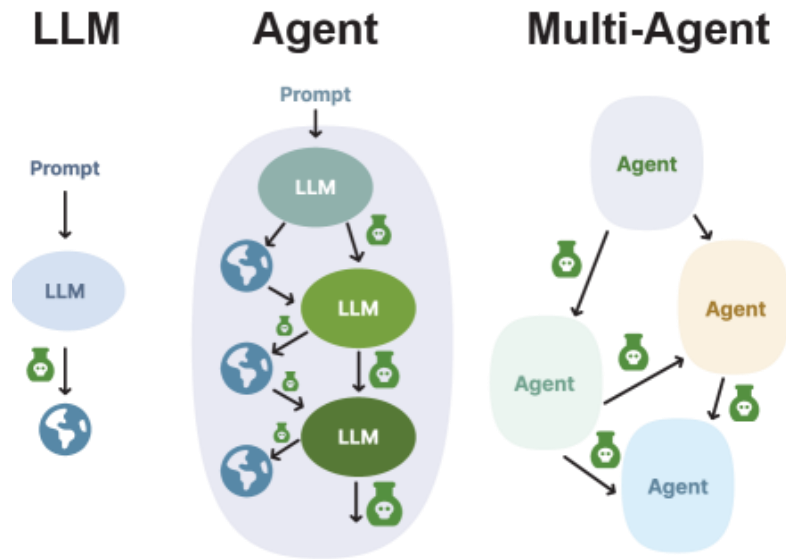
LLM agents: enabling LLMs to interact with the environment



LLM Agents in Diverse Environments



Multi-agent collaboration: division of labor for complex tasks



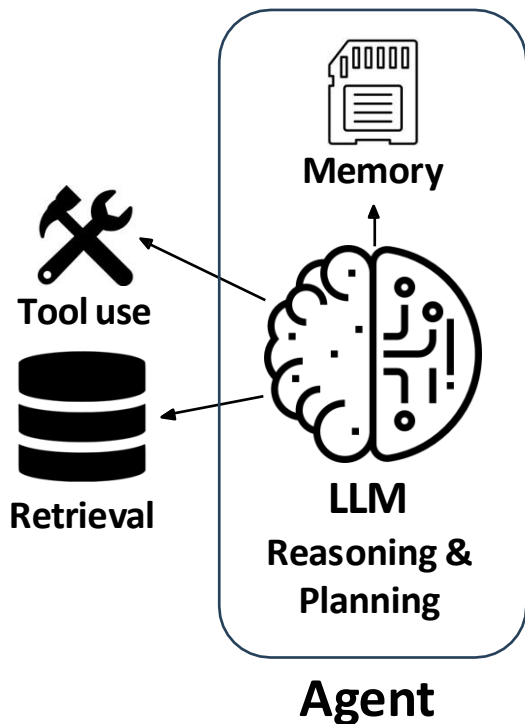
Specialized agents for different subtasks

Autogen, CrewAI, CAMEL, Mixture-of-Agents,...

Emergence of social behaviors with role-play LLMs

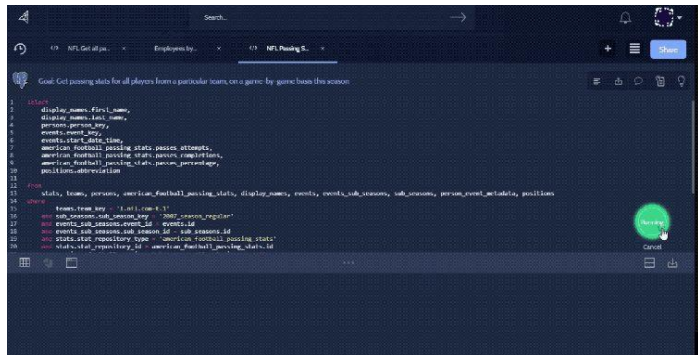
Generative agents, Project Sid,...

Why empowering LLMs with the agent framework



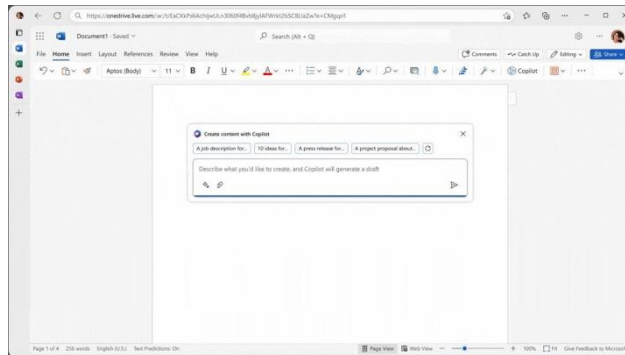
- Solving real-world tasks typically involves a trial-and-error process
- Leveraging external tools and retrieving from external knowledge expand LLM's information capabilities
- Agent workflow facilitates complex tasks
 - Allocation of subtasks to specialized tools
 - Multi-agent generation inspires better responses
 - Access to specialized evidence / data / inputs
 -

LLM agents transformed various applications



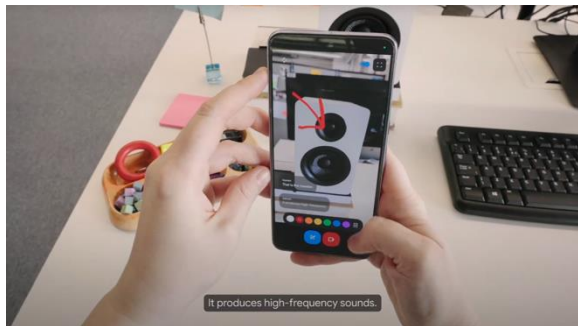
Code generation

Cursor, GitHub Copilot, Devin, Replit,...



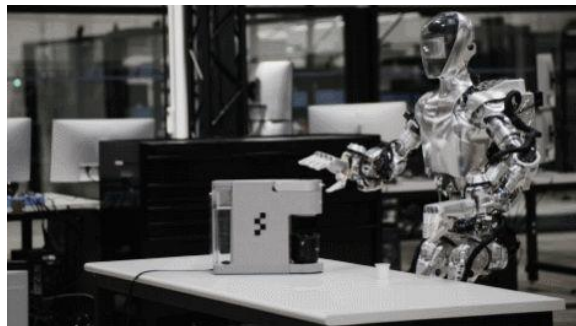
Workflow automation

Microsoft Copilot, Multi-On,...



Personal assistant

Google Astra, OpenAI GPT-4o,...



Robotics

Figure AI, Tesla Optimus,...

- Healthcare
- Education
- Law
- Finance
- Cybersecurity

...

LLM agents are improving rapidly (leaderboards!~)

GAIA (Mialon et al.)
huggingface.co/gaia-benchmark

Full is a large benchmark made of 2000 instances ([details](#))

Filters: Open Scaffold ▼ All Tags ▼

Model	% Resolved	Org	
✓ SWE-agent 1.0 (Claude 3.7 Sonnet)	33.83		Leaderboards
✓ OpenHands + CodeAct v2.1 (claude-3-5-sonnet-20241022)	29.38		BENCHMARKS
AutoCodeRover-v2.0 (Claude-3.5-Sonnet-20241022)	24.89		SWE-bench
✓ SWE-agent + Claude 3.5 Sonnet	18.13		SWE-bench Verified ↗
✓ SWE-agent + GPT 4 (1106)	12.47		SWE-bench Bash Only
✓ SWE-agent + GPT 4o (2024-05-13)	11.99		SWE-bench Multilingual
✓ SWE-agent + Claude 3 Opus	10.51		SWE-bench Multimodal
✓ RAG + Claude 3 Opus	3.79		SWE-bench Lite
			ABOUT

Results: Test

Agent name	Model family
JoinAI_V2.2	GPT 5, Gemini 3 Pro, DeepSeek 3.1, Qwen 3
Nemotron-Tool0r	Nemotron-Tool0rchestrator-8B, GPT-5, Claude Opus 4.1
Nemotron-Tool0r	Nemotron-Tool0rchestrator-8B, GPT-5, Claude Opus 4.1
SU Zero - Shugji	Self Consistency 35
JoinAI_V2.1	GPT, Gemini, DeepSeek, Qwen
ShawnAgent_v3.1	GPT5.2, Claude Sonnet 4.5, Gemini 3 Pro
HALO_V1217-1	

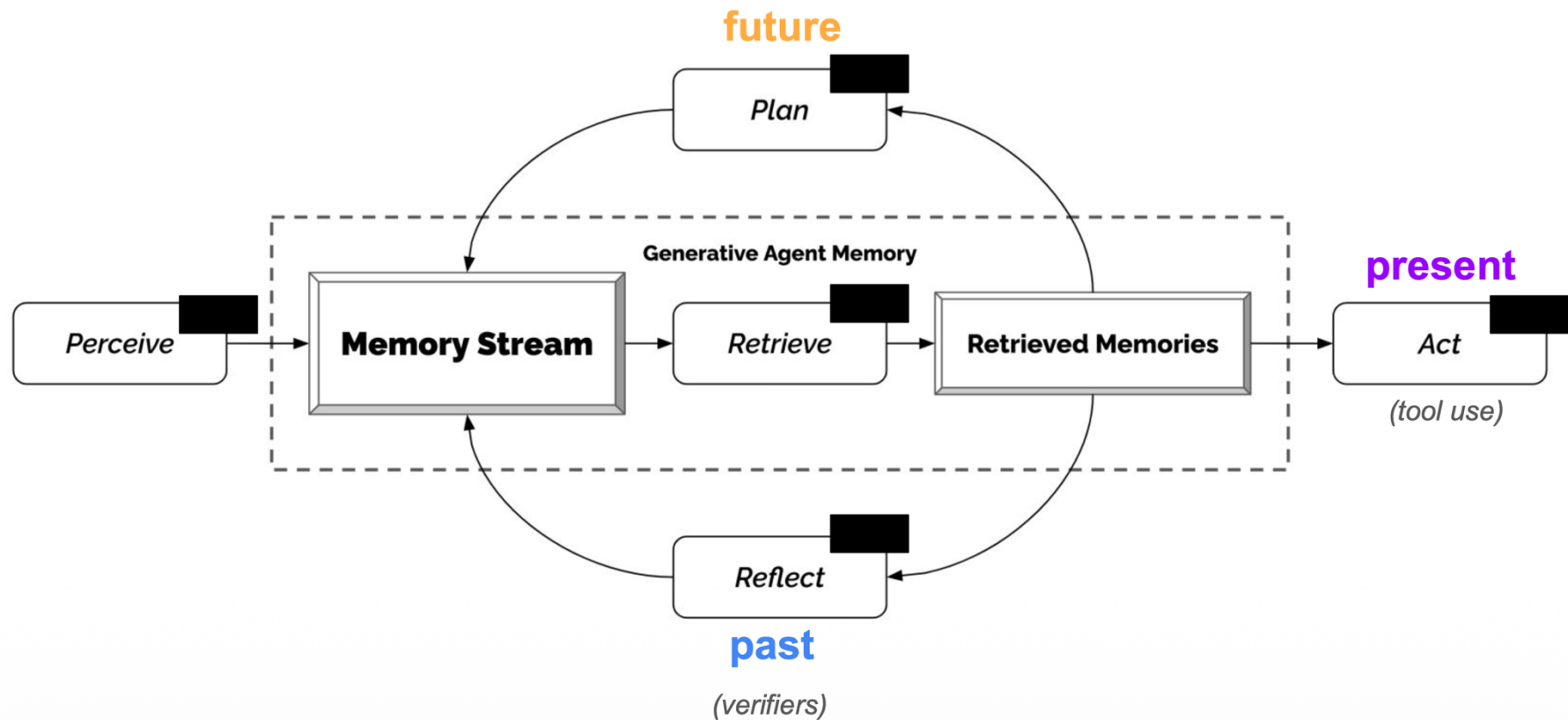
SWE-Bench (Jimenez*, Yang*, et al.) / swebench.com / Today's screenshot on Leaderboard!

F2	A	B	C	D	E	F	G	H	
1	a	Open?	Model Size (billion)	Model	Success Rate (%)	Result Source	Work	Traj	
2	01/2026	✗		OAgent	71.6	OAgent	OAgent	Link	
3	12/2025	✓	GPT-5	ColorBrowserAgent	71.2	ColorBrowserAgent	ColorBrowserAgent	Link	ite-sj
4	10/2025	✓	-	Claude Code + GBOX MCP	68	GBOX AI	GBOX AI	Link	
5	09/2025	✗	-	DeepSky Agent	66.9	Self-reported	DeepSky Agent	Link	
6	10/2025	✗		Narada AI	64.2	Self-reported	Narada AI	Link	
7	02/2025	✓	-	IBM CUGA	61.7	IBM CUGA	IBM CUGA	html + json	
8	01/2025	✗	-	OpenAI Operator	58.1	OpenAI CUA	OpenAI CUA	Link	Syste

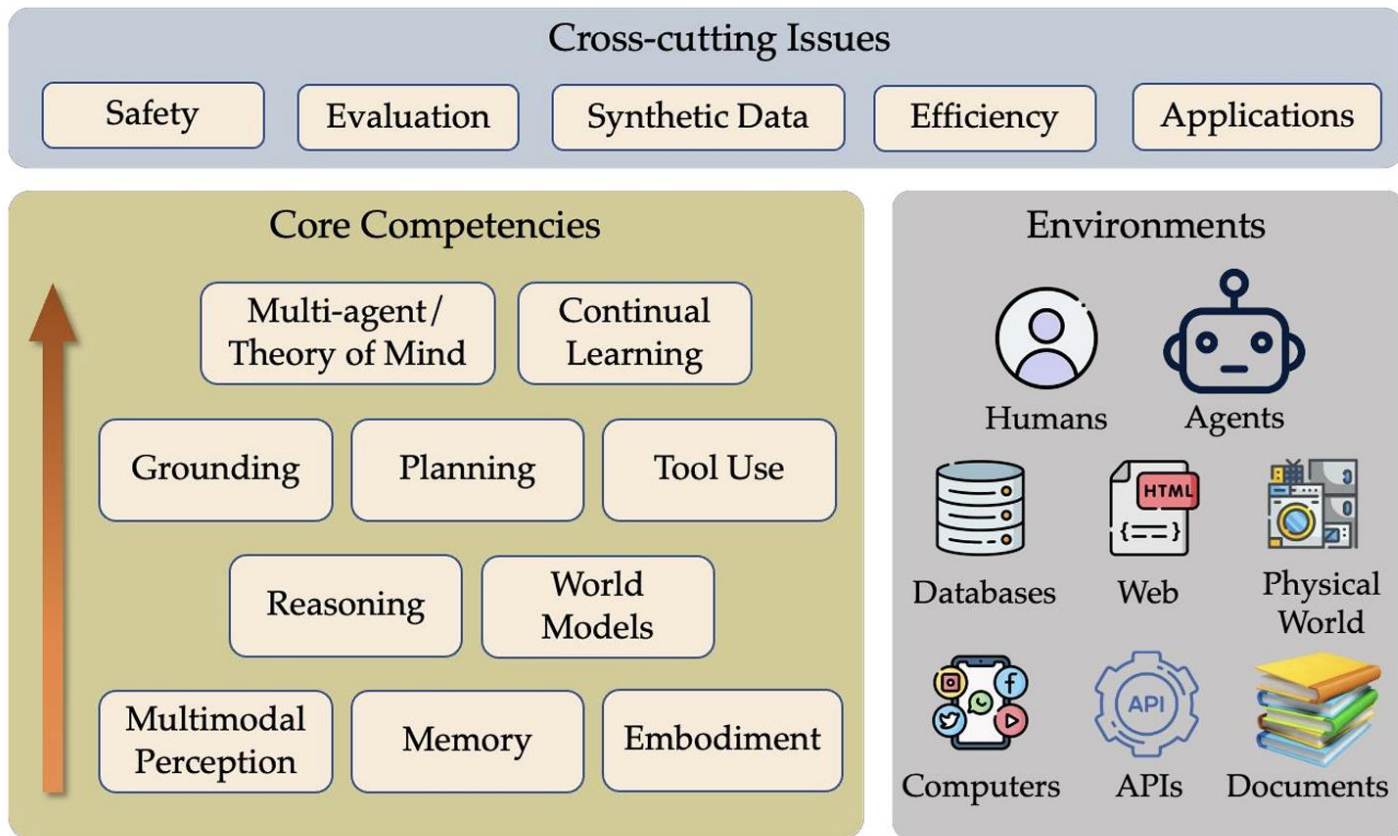
WebArena (Zhou et al.)
webarena.dev



Agent architecture



Topics We will cover in this course:



Topics covered in this course. ➔

- Applications
 - Software development
 - Workflow automation
 - Multimodal applications
 - Industrial applications like Healthcare, Legal, Fin, ...
- Model core capabilities
 - Reasoning
 - Planning
 - Multimodal understanding
- LLM agent frameworks
 - Workflow
 - Tool use
 - Retrieval-augmented generation
 - Multi-agent systems
- Safety and ethics

Potential Projects:

- Applications
 - Build LLM agents applications in specialized / novel domains
- Core Fundamentals
 - Enhance core agent capabilities (memory, planning, tool use, alignments, efficiency, ...)
 - Enhance decentralized multi-agent systems
- Benchmarks / Build Novel Frameworks
 - Create and improve benchmarks for Evaluating LLM agents
 - Reimplement or Build novel frameworks for agent workflow
- Safety and ethics
 - Reveal safety concerns in deployment (misuse, privacy, etc.)
 - Defense safety concerns

Challenges for LLM agent deployment in the wild

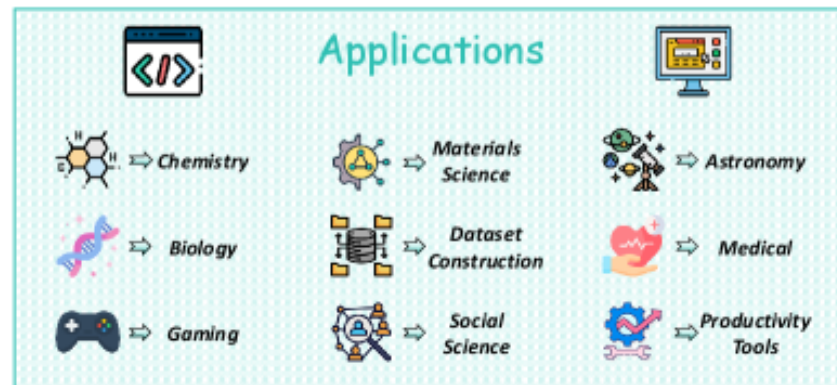
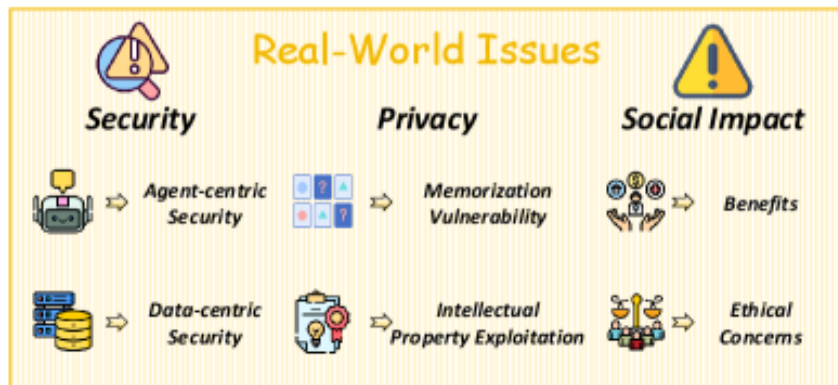
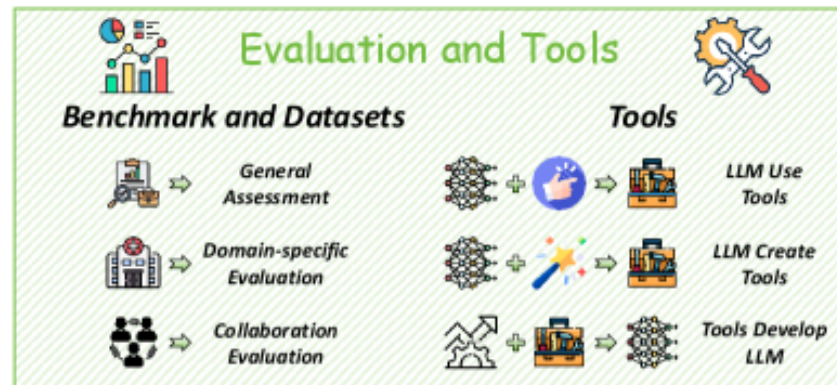
- Reasoning and planning
 - LLM agents tend to make mistakes when performing complex tasks end-to-end
- Embodiment and learning from environment feedback
 - LLM agents are not yet efficient at recovering from mistakes for long-horizon tasks
 - Continuous learning, self-improvement
 - Multimodal understanding, grounding and world models
- Multi-agent learning, theory of mind
- Safety and privacy
 - LLMs are susceptible to adversarial attacks, can emit harmful messages and leak private data
- Human-agent interaction, ethics
 - How to effectively control the LLM agent behavior, and design the interaction mode between humans and LLM agents

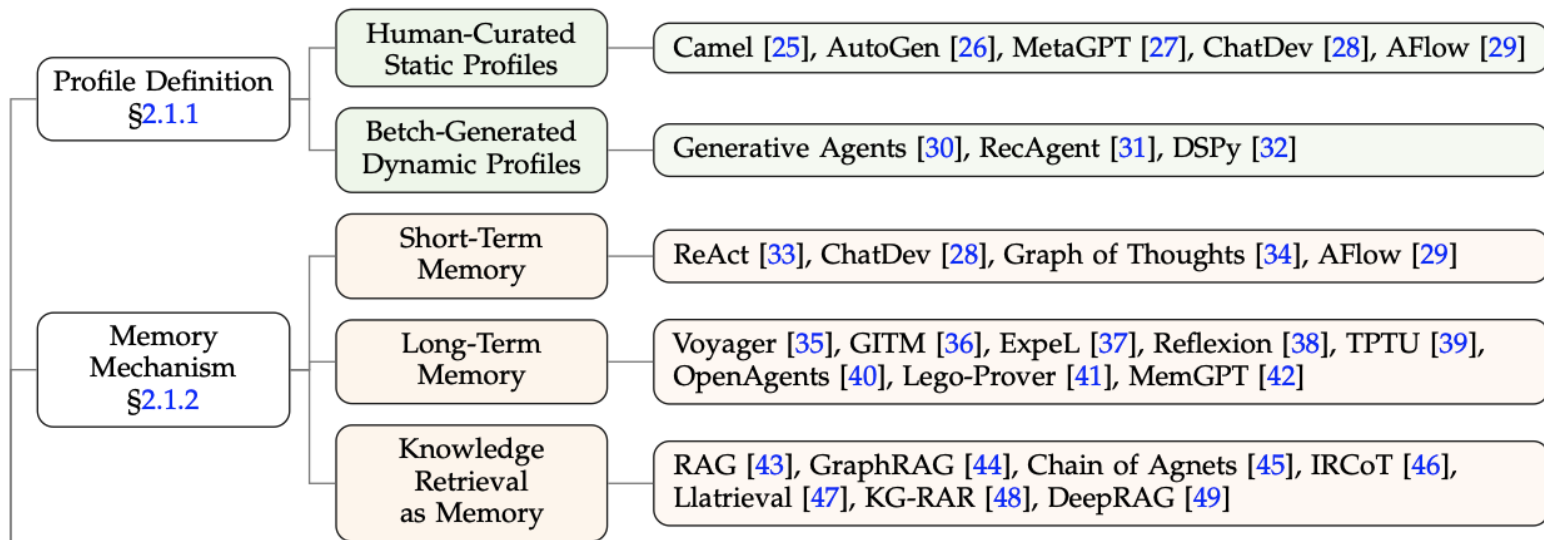
Large Language Model Agent: A Survey on Methodology, Applications and Challenges

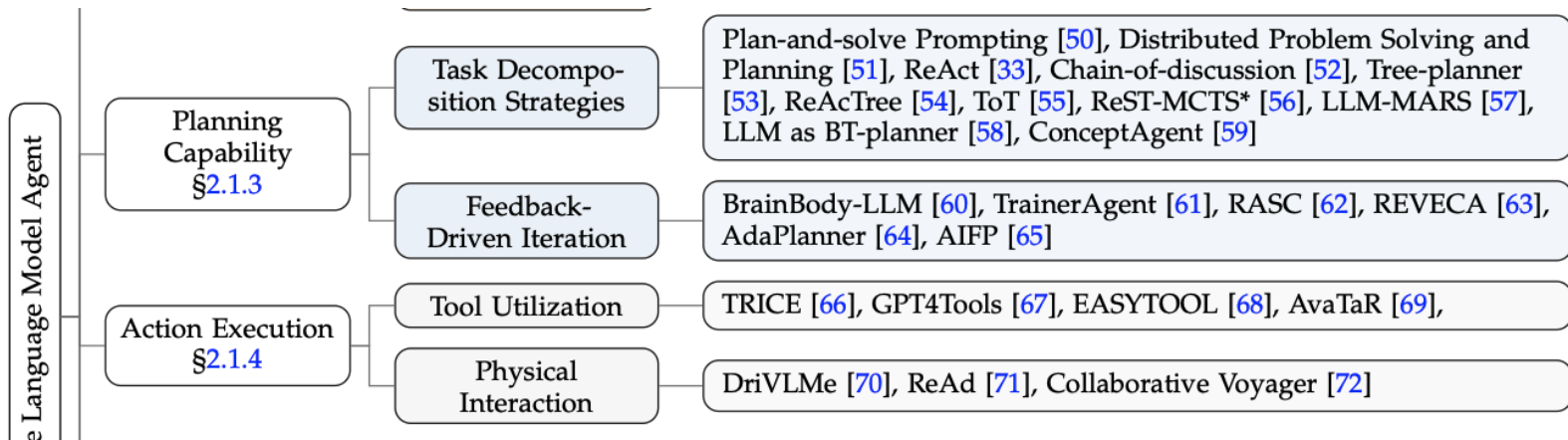
Junyu Luo, Weizhi Zhang, Ye Yuan, Yusheng Zhao, Junwei Yang, Yiyang Gu, Bohan Wu, Binqi Chen, Ziyue Qiao, Qingqing Long, Rongcheng Tu, Xiao Luo, Wei Ju, Zhiping Xiao, Yifan Wang, Meng Xiao, Chenwu Liu, Jingyang Yuan, Shichang Zhang, Yiqiao Jin, Fan Zhang, Xian Wu, Hanqing Zhao, Dacheng Tao, *Fellow, IEEE*, Philip S. Yu, *Fellow, IEEE* and Ming Zhang

Abstract—The era of intelligent agents is upon us, driven by revolutionary advancements in large language models. Large Language Model (LLM) agents, with goal-driven behaviors and dynamic adaptation capabilities, potentially represent a critical pathway toward artificial general intelligence. This survey systematically deconstructs LLM agent systems through a methodology-centered taxonomy, linking architectural foundations, collaboration mechanisms, and evolutionary pathways. We unify fragmented research threads by revealing fundamental connections between agent design principles and their emergent behaviors in complex environments. Our work provides a unified architectural perspective, examining how agents are constructed, how they collaborate, and how they evolve over time, while also addressing evaluation methodologies, tool applications, practical challenges, and diverse application domains. By surveying the latest developments in this rapidly evolving field, we offer researchers a structured taxonomy for understanding LLM agents and identify promising directions for future research. The collection is available at <https://github.com/luo-junyu/Awesome-Agent-Papers>.

Index Terms—Large language model, LLM agent, AI agent, intelligent agent, multi-agent system, LLM, literature survey







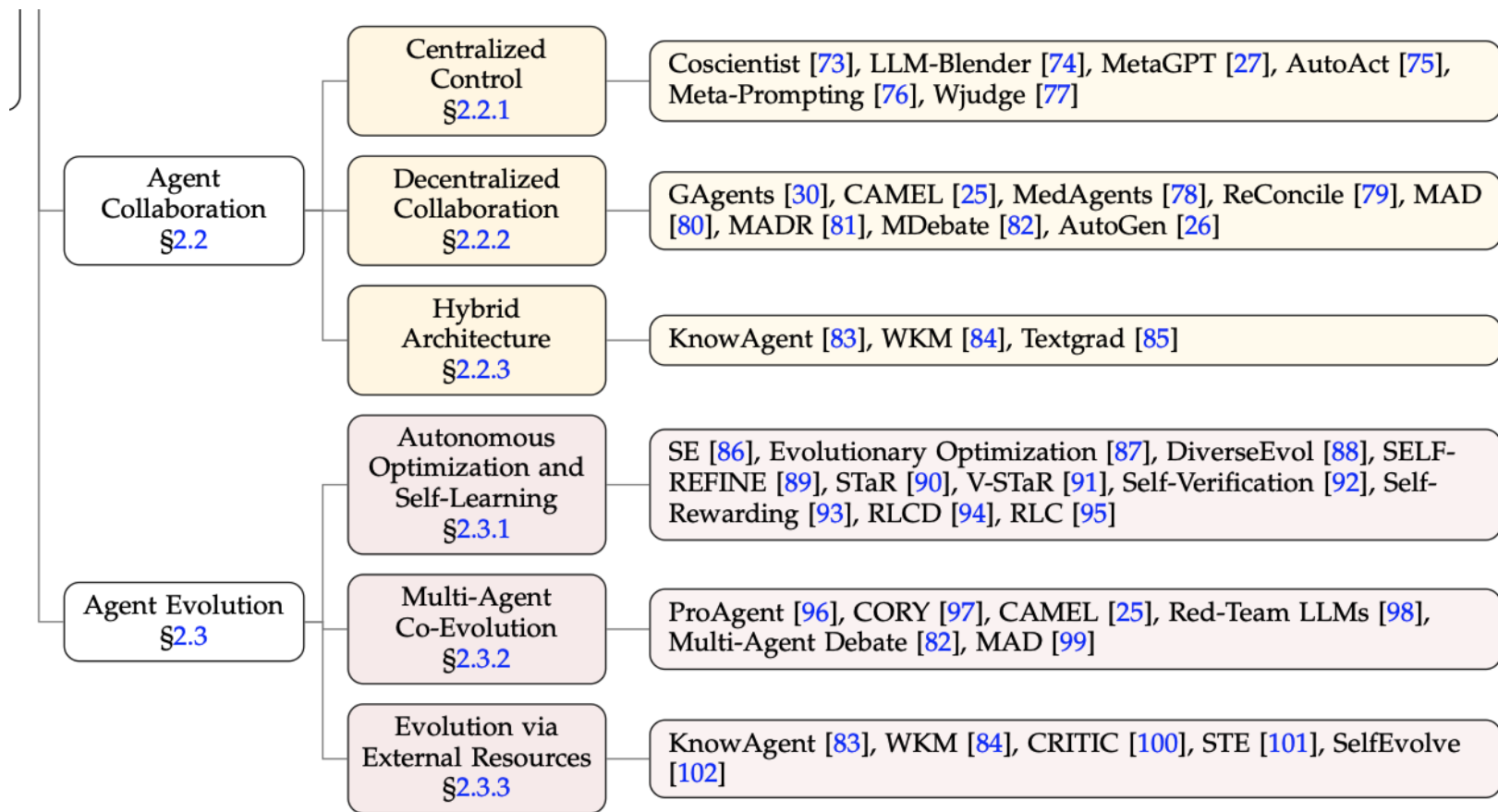


TABLE 2: A summary of agent evolution methods.

Category	Method	Key Contribution
Self-Supervised Learning	SE [86] Evolutionary Optimization [87] DiverseEvol [88]	Adaptive token masking for pretraining Efficient model merging and adaptation Improved instruction tuning via diverse data
Self-Reflection & Self-Correction	SELF-REFINE [89] STaR [90] V-STaR [91] Self-Verification [92]	Iterative self-feedback for refinement Bootstrapping reasoning with few rationales Training a verifier using DPO Backward verification for correction
Self-Rewarding & RL	Self-Rewarding [93] RLCD [94] RLC [95]	LLM-as-a-Judge for self-rewarding Contrastive distillation for alignment Evaluation-generation gap for optimization
Cooperative Co-Evolution	ProAgent [96] CORY [97] CAMEL [25]	Intent inference for teamwork Multi-agent RL fine-tuning Role-playing framework for cooperation
Competitive Co-Evolution	Red-Team LLMs [98] Multi-Agent Debate [82] MAD [99]	Adversarial robustness training Iterative critique for refinement Debate-driven divergent thinking
Knowledge-Enhanced Evolution	KnowAgent [83] WKM [84]	Action knowledge for planning Synthesizing prior and dynamic knowledge
Feedback-Driven Evolution	CRITIC [100] STE [101] SelfEvolve [102]	Tool-assisted self-correction Simulated trial-and-error for tool learning Automated debugging and refinement

Real-world Issues


Agent-centric
Security

   
Adversarial
Attacks Jailbreak
Attacks Backdoor
Attacks Model
Collaboration


Data-centric
Security

 
External Data
Attacks Interaction
Attacks

Security


Memorization
Vulnerability

  
Data
Extraction
Attacks Member
Inference
Attacks Attribute
Inference
Attacks


Intellectual Property
Exploitation

 
Model Stealing
Attacks Prompt Stealing
Attacks

Privacy


Benefits

  
Automation
Enhancement Job Creation
and Workforce
Transformation Enhance
Information
Distribution


Ethical Concerns

  
Bias and
Discrimination Accountability

**Social
Impact**

TABLE 3: Summary of agent-centric attacks and defense in LLM agents.

Reference	Description
Adversarial Attacks and Defense	
Mo et al. [177]	Attack: Adversarial attack benchmark
AgentDojo [178]	Attack: Adversarial attack framework
ARE [179]	Attack: Adversarial attack evaluation for multimodal agents
GIGA [181]	Attack: Generalizable infectious gradient attacks
CheatAgent [180]	Attack: Adversarial attack agent for recommender systems
LLAMOS [182]	Defense: Purifying adversarial attack input
Chern et al. [183]	Defense: Defense via multi-agent debate
Jailbreaking Attacks and Defense	
RLTA [184]	Attack: Produce jailbreaking prompts via reinforcement learning
Atlas [185]	Attack: Jailbreaks text-to-image models with safety filters
RLbreaker [186]	Attack: Model jailbreaking as a search problem
PathSeeker [187]	Attack: Use multi-agent reinforcement learning to jailbreak
AutoDefense [188]	Defense: Multi-agent defense to filter harmful responses
Guardians [189]	Defense: Detect rogue agents to counter jailbreaking attacks.
ShieldLearner [190]	Defense: Learn attack jailbreaking patterns.
Backdoor Attacks and Defense	
DemonAgent [191]	Attack: Encrypted muti-backdoor implantation attack
Yang et al. [192]	Attack: Backdoor attacks evaluations on LLM-based agents
BadAgent [193]	Attack: Inputs or environment cues as backdoors
BadJudge [194]	Attack: Backdoor to the LLM-as-a-judge agent system
DarkMind [195]	Attack: latent backdoor attack to customized LLM agents
Agent Collaboration Attacks and Defense	
CORBA [196]	Attack: Multi-agent attack via multi-agent

TABLE 4: Summary of data-centric attack and defense in LLM agents.

Reference	Description
External Data Attacks and Security	
Li et al. [204]	Attack: Malicious prefix injection
Psysafe [201]	Attack: A dark psychological injection benchmark
Tian et al. [210]	Attack: Guide agents into specific role-playing states
InjectAgent [205]	Attack: A prompting injection benchmark
Agentdojo [203]	Attack: A user injection benchmark
AgentPoison [216]	Attack: Poisoning samples in knowledge databases
Nakash et al. [215]	Attack: Indirect prompt injection through FITD attack
WIPI [214]	Attack: control agents through a public web page
ASB [176]	Attack: A multi-type attack benchmark
AgentHarm [223]	Attack: A multi-type attack benchmark
Mantis [206]	Defense: Hacking back to attackers
Chern et al. [183]	Defense: Employ multi-agent debate to verify external knowledge
RTBAS [208]	Defense: Check every step of agent information flow
TaskShield [209]	Defense: Check every step of agent process
Zhang et al. [201]	Defense: Doctor and police agents guard the healthy psychology
Interaction Attacks and Security	
Wang et al. [217]	Attack: Private memory extraction attack
CORBA [196]	Attack: Disrupt the communications among agents
AgentSmith [220]	Attack: Poison one agent to infectious other agents
Lee et al. [221]	Attack: Conduct injections to self-replicate among agents
He et al. [197]	Attack: Inject semantic disruptions to agent communications
BlockAgents [222]	Defense: Incorporate blockchain and PoT against byzantine attacks
Abdelnabi et al. [207]	Defense: A multi-layer agent firewall

TABLE 7: Overview of Applications in LLM Agents.

Method	Domain	Core Idea
Scientific Discovery		
SciAgents [266]	General Sciences	Collaborative hypothesis generation
Curie [267]	General Sciences	Automated experimentation
ChemCrow [269]	Chemistry	Tool-augmented synthesis planning
AtomAgents [270]	Materials Science	Physics-aware alloy design
D. Kostunin et al [271]	Astronomy	Telescope configuration management
BioDiscoveryAgent [273]	Biology	Genetic perturbation design
GeneAgent [274]	Biology	Self-verifying gene association discovery
RiGPS [275]	Biology	Biomarker identification
BioRAG [211]	Biology	Biology-focused retrieval augmentation
PathGen-1.6M [276]	Medical Dataset	Pathology image dataset generation
KALIN [277]	Biology Dataset	Scientific question corpus generation
GeneSUM [278]	Biology Dataset	Gene function knowledge maintenance
AgentHospital [281]	Medical	Virtual hospital simulation
ClinicalLab [282]	Medical	Multi-department diagnostics
AlPatient [283]	Medical	Patient simulation
CXR-Agent [284]	Medical	Chest X-ray interpretation
MedRAX [285]	Medical	Multimodal medical reasoning
Gaming		
ReAct [33]	Game Playing	Reasoning and acting in text environments
Voyager [35]	Game Playing	Lifelong learning in Minecraft
ChessGPT [287]	Game Playing	Chess gameplay evaluation
GLAM [288]	Game Playing	Reinforcement learning in text environments
CALYPSO [289]	Game Generation	Narrative generation for D&D
GameGPT [290]	Game Generation	Automated game development
Sun et al. [291]	Game Generation	Interactive storytelling experience

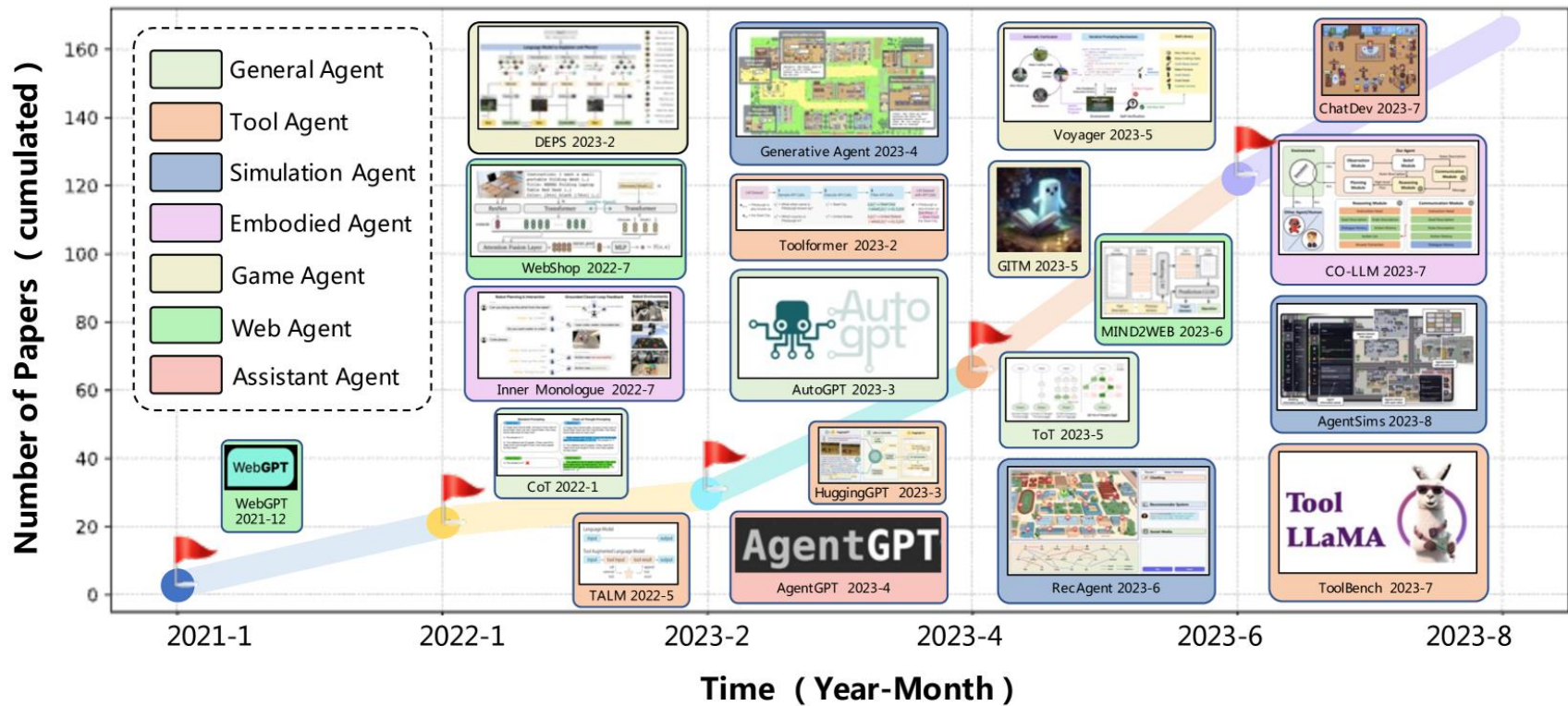
A Survey on Large Language Model based Autonomous Agents

Lei Wang, Chen Ma*, Xueyang Feng*, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhi-Yuan Chen, Jiakai Tang, Xu Chen(✉), Yankai Lin(✉), Wayne Xin Zhao, Zhewei Wei, Ji-Rong Wen

Gaoling School of Artificial Intelligence, Renmin University of China, Beijing, 100872, China

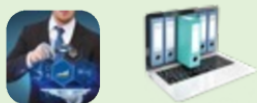
© Higher Education Press 2025

Abstract Autonomous agents have long been a research focus in academic and industry communities. In this survey, we first review the existing LLM-based autonomous agents. Based on the previous studies, we also present several challenges





Profile



Profile Contents

- Demographic Information
- Personality Information
- Social Information

Generation Strategy

- Handcrafting Method
- LLM-Generation Method
- Dataset Alignment Method

Memory



Memory Structure

- Unified Memory
- Hybrid Memory

Memory Formats

- Languages
- Databases
- Embeddings
- Lists

Memory Operation

- Memory Reading
- Memory Writing
- Memory Reflection

Planning



Planning w/o Feedback

- Single-path Reasoning
- Multi-path Reasoning
- External Planner

Planning w/ Feedback

- Environment Feedback
- Human Feedback
- Model Feedback

Action



Action Target

- Task Completion
- Exploration
- Communication

Action Production

- Memory Recollection
- Plan Following

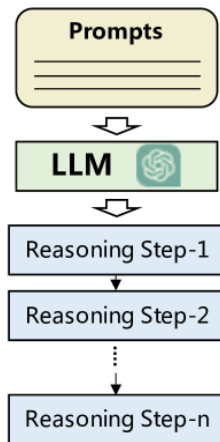
Action Space

- Tools
- Self-Knowledge

Action Impact

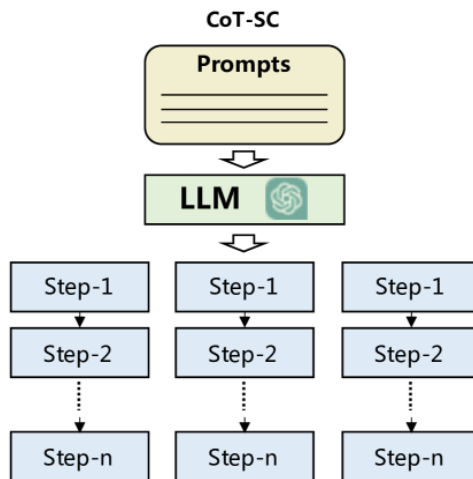
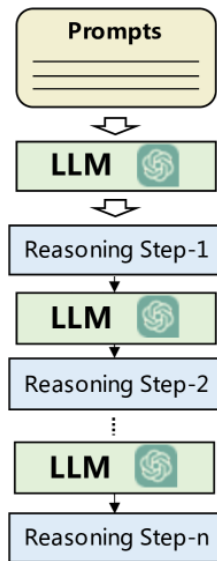
- Environments
- New Actions
- Internal States

CoT , Zero-shot Cot



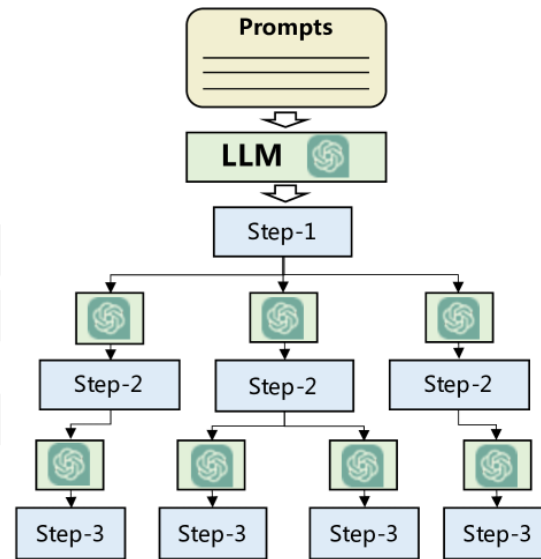
Single-Path Reasoning

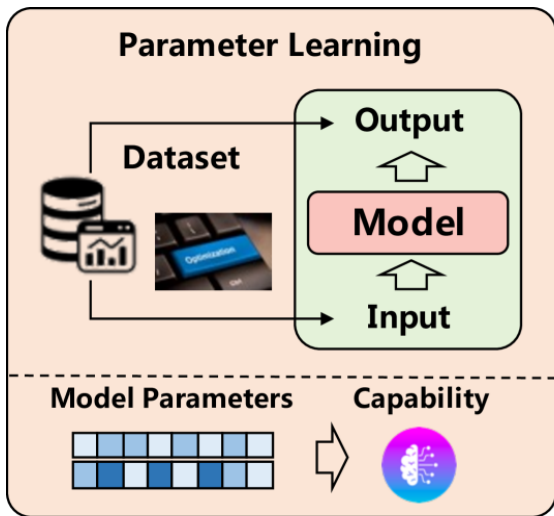
ReWOO , HuggingGPT



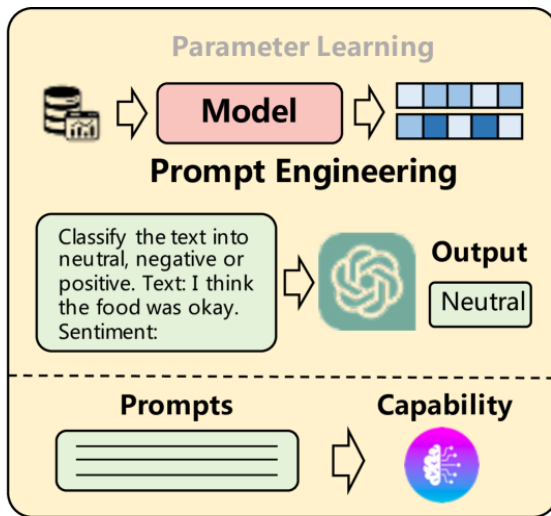
Multi-Path Reasoning

ToT , LMZSP , RAP

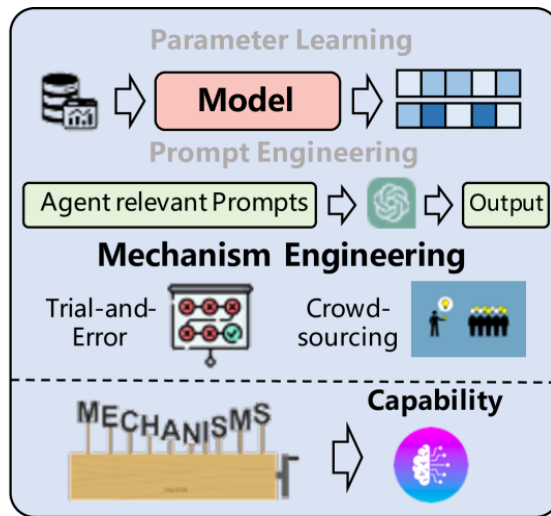




The era of machine learning



The era of large language model



The era of agent

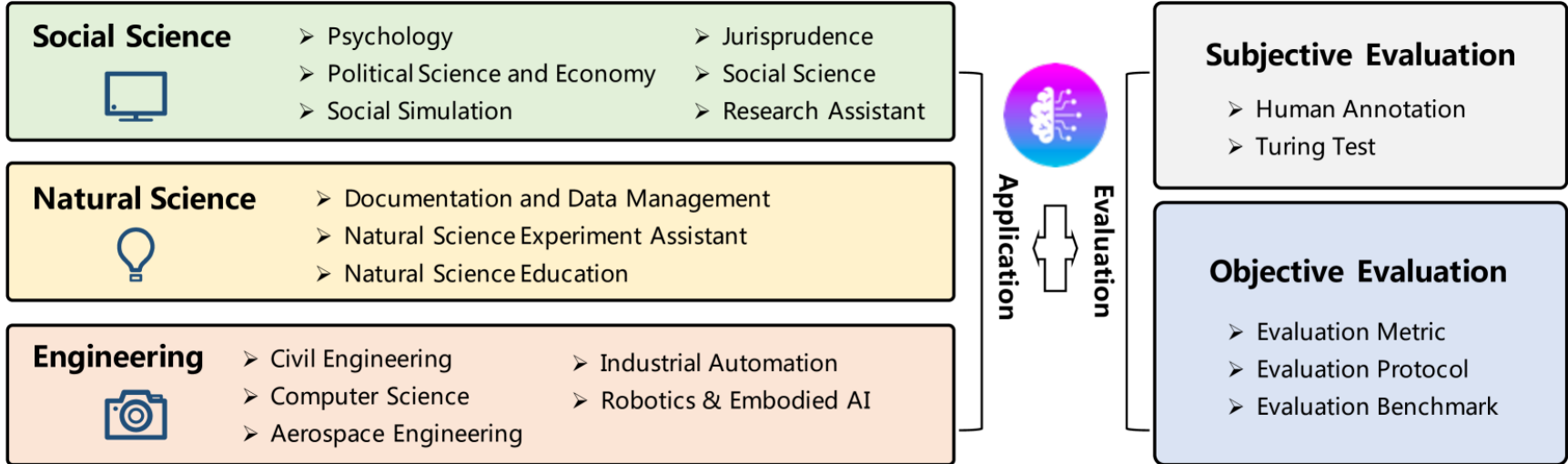


Table 2 Representative applications of LLM-based autonomous agents.

	Domain	Work
Social Science	Psychology	TE [101], Akata et al. [102], Ziems et al. [104], Ma et al. [103]
	Political Science and Economy	Argyle et al. [29], Horton [105], Ziems et al. [104]
	Social Simulation	Social Simulacra [80], Generative Agents [20], SocialAI School [108], AgentSims [34], S ³ [78], Williams et al. [109], Li et al. [106], Chao et al. [107]
	Jurisprudence	ChatLaw [111], Blind Judgement [112]
	Research Assistant	Ziems et al. [104], Bail et al. [113]
	Documentation and Data Management	ChemCrow [76], ChatMOF [115], Boiko et al. [114]

	Research Assistant	Ziems et al. [104], Bail et al. [113]
Natural Science	Documentation and Data Management	ChemCrow [76], ChatMOF [115], Boiko et al. [114]
	Experiment Assistant	ChemCrow [76], Boiko et al. [114], Grossmann et al. [121]
	Natural Science Education	ChemCrow [76], CodeHelp [119], Boiko et al. [114], MathAgent [116], Drori et al. [117], EduChat [87], FreeText [120]
Engineering	CS & SE	RestGPT [71], Self-collaboration [24], SQL-PALM [89], RAH [91], D-Bot [122], RecMind [53], ChatEDA [123], InteRecAgent [124], PentestGPT [125], CodeHelp [119], SmolModels [126], DemoGPT [127], GPTEngineer [128]
	Industrial Automation	GPT4IA [129], IELLM [130]
	Robotics & Embodied AI	ProAgent [131], LLM4RL [132], PET [133], REMEMBERER [134], DEPS [33], Unified Agent [135], SayCan [79], TidyBot [136], RoCo [92], SayPlan [31], TaPA [137], Dasgupta et al. [138], DECKARD [139], Dialogue shaping [140]

Table 3 For subjective evaluation, we use ① and ② to represent human annotation and the Turing test, respectively. For objective evaluation, we use ①, ②, ③, and ④ to represent real-world simulation, social evaluation, multi-task evaluation, and software testing, respectively. “✓” indicates that the evaluations are based on benchmarks.

Model	Subjective	Objective	Benchmark	Time
WebShop [86]	-	① ③	✓	07/2022
Social Simulacra [80]	①	②	-	08/2022
TE [101]	-	②	-	08/2022
LIBRO [159]	-	④	-	09/2022
ReAct [60]	-	①	✓	10/2022
Argyle et al. [29]	②	② ③	-	02/2023
DEPS [33]	-	①	✓	02/2023
Jalil et al. [160]	-	④	-	02/2023
Reflexion [12]	-	① ③	-	03/2023
IGLU [161]	-	①	✓	04/2023
Generative Agents [20]	①	-	-	04/2023
ToolBench [151]	-	③	✓	04/2023
GITM [16]	-	①	✓	05/2023
Toolformer [162]	-	④	-	05/2023

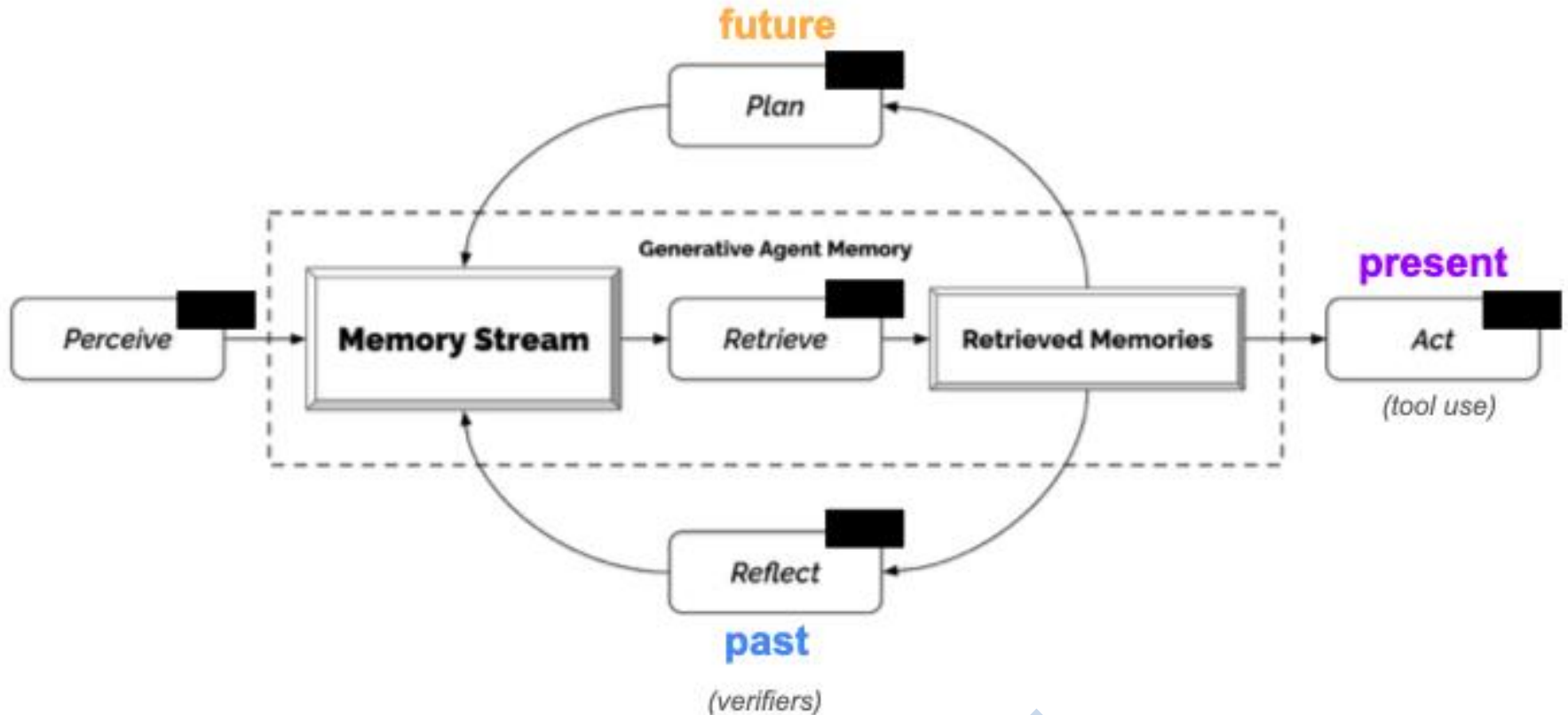
ToolBench [151]	-	③	✓	04/2023
GITM [16]	-	①	✓	05/2023
Two-Failures [162]	-	③	-	05/2023
Voyager [38]	-	①	✓	05/2023
SocKET [163]	-	② ③	✓	05/2023
MobileEnv [164]	-	① ③	✓	05/2023
Clembench [165]	-	① ③	✓	05/2023
Dialop [166]	-	③	✓	06/2023
Feldt et al. [167]	-	④	-	06/2023
CO-LLM [22]	①	①	-	07/2023
Tachikuma [168]	①	① ③	✓	07/2023
RocoBench [92]	-	① ③	✓	07/2023
AgentSims [34]	-	②	-	08/2023
AgentBench [169]	-	③	✓	08/2023
BOLAA [170]	-	③	✓	08/2023
Gentopia [171]	-	③	✓	08/2023
EmotionBench [172]	①	-	✓	08/2023
PTB [125]	-	④	-	08/2023

Many new surveys and new frameworks!



Another Framework:

Agent architecture



Credit: from Stanford Prof. Percy Liang

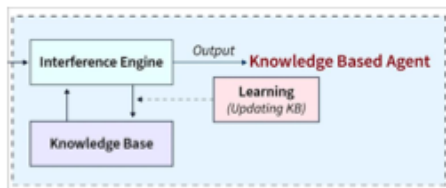
What is “agent”?



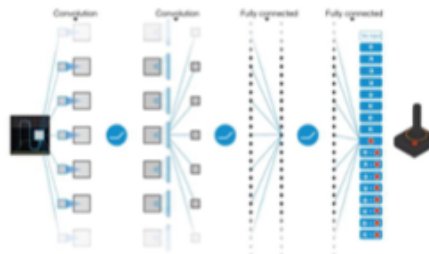
- An “intelligent” system that interacts with some “environment”
 - Physical environments: robot, autonomous car, ...
 - Digital environments: DQN for Atari, Siri, AlphaGo, ...
 - Humans as environments: chatbot
- Define “agent” by defining “intelligent” and “environment”
 - It changes over time!
 - Exercise question: how would you define “intelligent”?

Another Framework:

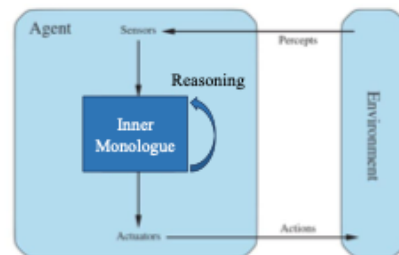
Evolution of AI agents



Logical Agent



Neural Agent



Language Agent

Expressiveness	Low bounded by the logical language	Medium anything a (small-ish) NN can encode	High almost anything, esp. verbalizable parts of the world
Reasoning	Logical inferences sound, explicit, rigid	Parametric inferences stochastic, implicit, rigid	Language-based inferences fuzzy, semi-explicit, flexible
Adaptivity	Low bounded by knowledge curation	Medium data-driven but sample inefficient	High strong prior from LLMs + language use

LLM Agent Frameworks & Benchmarks

LIBRARIES / FRAMEWORKS

LangChain (Feb)

- Enables chaining LLMs with various tools and multi-step workflows.
- Various tools like APIs, databases, and external data sources.
- Memory mgmt, allowing context retention across multiple interactions.

AutoGPT (Mar)

- Automates tasks with autonomous agents.
- Uses a feedback loop to refine outputs based on goals and constraints.
- Unlike LangChain, emphasizes autonomous decision-making over structured workflow chaining.

AutoGen (Sept)

- Multi-agent framework for building workflows with AI agents.
- AutoGen agents can work together, integrating LLMs, tools, and human inputs.
- Unlike LangChain and AutoGPT, emphasize multi-agent interaction and human-AI collab

Crew.ai (Dec)

- Collaborative agent teams with specific roles and goals.
- Sequential and hierarchical processes.
- Versatile tools with error handling and caching capabilities.
- Allows human oversight & interaction

2022

2023

2024

BENCHMARKS

ToolBench (May)

- Evaluate tool use with diverse real-world tasks
- 8 tasks, e.g.: Open Weather, Trip booking, Google Sheets
- Can boost open-source LLMs to 90% success rate, matching GPT-4 in 4 out of 8 tasks

AgentBench (Aug)

- 8 environments:
- operating system
 - database
 - knowledge graph
 - digital card game
 - lateral thinking puzzles
 - house-holding
 - web shopping
 - web browsing

MLAgentBench (Oct)

- 13 tasks for ML experimentation, from CIFAR-10 to BabyLM.
- Tasks include file operations, run code, output inspection.
- Best is Claude v3 Opus 37.5% avg success rate
- Challenges: long-term planning, hallucinations

GAIA (Nov)

- Q&A: need reasoning, multi-modality, tools.
- Humans: 92% vs. 15% for GPT-4 with plugins.
- 466 questions; 166 with detailed traces, 300 retained for leaderboard.
- Questions have

Crew.ai (Dec)

- Collaborative agent teams with specific roles and goals.
- Sequential and hierarchical processes.
- Versatile tools with error handling and caching capabilities.
- Allows human oversight & interaction

LangGraph (Jan)

- Graph-based: agent workflows as nodes and edges
- Stateful design
- Supports human-agent collaboration
- Real-time streaming
- Allows granular control

LlamaIndex Workflows (Aug)

- Event-driven architecture
- Provides state management and enables cyclical flows
- Supports tools like Arize Phoenix for debugging

TapeAgents (Oct)

- Single unifying abstraction (the "tape") which is both a log of events and the state of the system
- Enables complex agent optimization such as prompt tuning and distillation from complex teacher to simpler student

2024

BENCHMARKS**GAIA** (Nov)

- Q&A: need reasoning, multi-modality, tools.
- Humans: 92% vs. 15% for GPT-4 with plugins.
- 466 questions; 166 with detailed traces, 300 retained for leaderboard.
- Questions have unambiguous answer.

SWE-Bench (Apr)

- Evaluate AI agents on real-world software engineering tasks
- 2,294 problems from real GitHub issues and PR across 12 popular Python repositories
- Code generation, bug fixing, design
- Evals on correctness, efficiency, collab

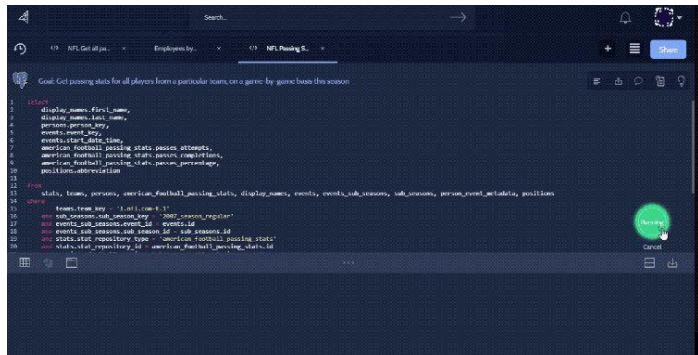
τ-Bench (Jun)

- Emulate conversations between a LLM user and a LLM agent provided with domain-specific API tools and policy guidelines
- 175 tasks from retail and airline domains
- Top models still at sub-par performance

InsightBench (Oct)

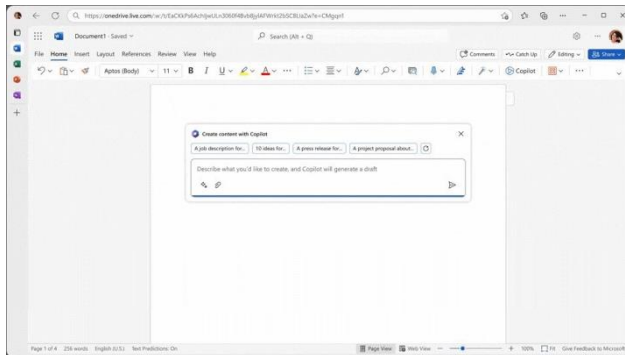
- Evaluate agents on end-to-end data science workflows, measuring cross-domain generalization
- Task planning, execution, reasoning
- Incomplete data & ambiguous goals

Recap: LLM agents transformed various applications



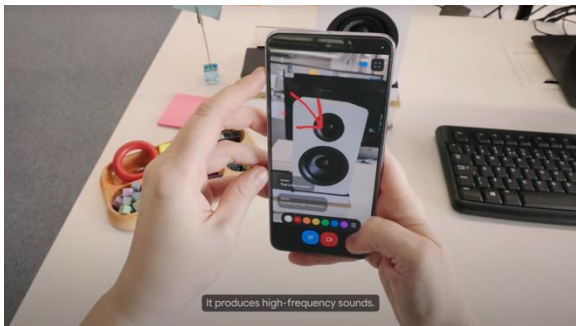
Code generation

Cursor, GitHub Copilot, Devin, Replit,...



Workflow automation

Microsoft Copilot, Multi-On,...



Personal assistant

Google Astra, OpenAI GPT-4o,...



Robotics

Figure AI, Tesla Optimus,...

- Healthcare
- Education
- Law
- Finance
- Cybersecurity

...

Please start to build one agent by
yourself / your team!

<https://github.com/openclaw/openclaw>

 **openclaw** Public

Watch 839 Fork 24.2k Starred 157k

main Go to file + Code

 **steipete** chore: prepare 2026.2.2 release ✓ 1c4db91 · 48 minutes ago ⌚ 8,795 Commits

📁 .agent/workflows	chore: Run pnpm format:fix.	3 days ago
📁 .github	fix: CI: We no longer need to test the ts...	6 hours ago
📁 .pi	chore: fix Pi prompt template argument...	2 days ago
📁 Swabble	refactor: rename to openclaw	5 days ago

About

Your own personal AI assistant. Any OS.
Any Platform. The lobster way. 🦞

[openclaw.ai](#)

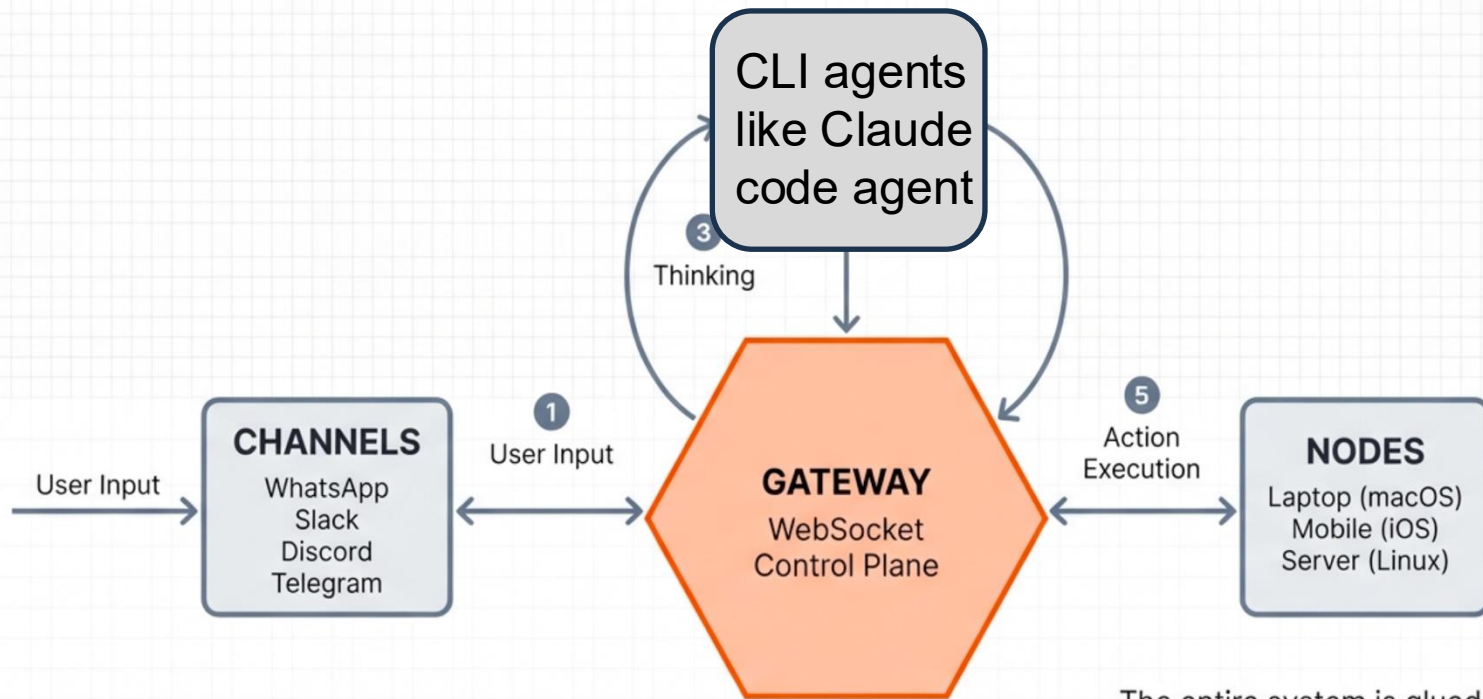
ai personal assistant own-your-data

crustacean molty openclaw

📖 Readme

📄 MIT license

Openclaw / moltbot



The entire system is glued together by a unified WebSocket control plane.

<https://github.com/block/goose>

[Code](#) [Issues 249](#) [Pull requests 89](#) [Discussions](#) [Actions](#) [Projects 2](#) [Security](#) [Insights](#)

 **goose** Public Watch 170 Fork 2.7k Starred 29.8k

main Code

 **Isytj0413** feat: pass env to shell comm... 3e38c30 · 8 minutes ago 3,377 Commits

.devcontainer	fix(devcontainer): install protoc to fix ...	7 months ago
.github	feat: ask ai discord bot (#6842)	20 hours ago
.husky	Remove a bit from pre-commit that hu...	6 months ago
.intersect	[FEAT] Introduce PR level security sca...	last year

About

an open source, extensible AI agent that goes beyond code suggestions - install, execute, edit, and test with any LLM

block.github.io/goose/

mcp

Readme

Apache-2.0 license